

EX NO:1	Butterworth IIR Filters
DATE:	1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

Code:

```

clc;
clear;
close all;

% EX NO: 1.1
% Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB

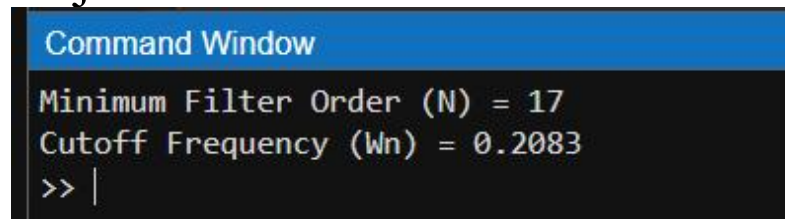
Fs = 10000;    % Sampling Frequency (Hz)
Fp = 1000;     % Passband Frequency (Hz)
Fst = 1500;    % Stopband Frequency (Hz)

% Calculate minimum filter order
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);
% Design Butterworth Low-Pass Filter
[b, a] = butter(N, Wn);
% Magnitude and Phase Response
figure;
freqz(b, a);
title('Frequency Response');
% Impulse Response
figure;
impz(b, a, 50);
title('Impulse Response');
% Step Response
figure;
stepz(b, a);
title('Step Response');
% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot');
% Display Results
fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);

```

Outputs:

Objective 1:



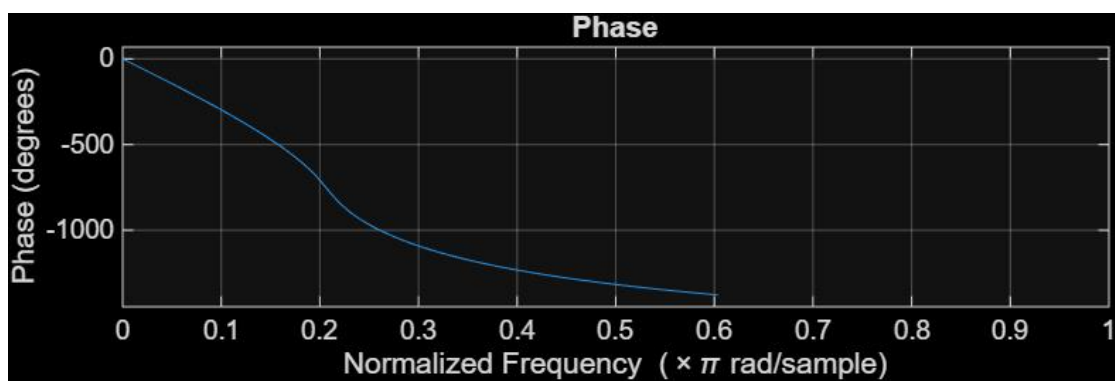
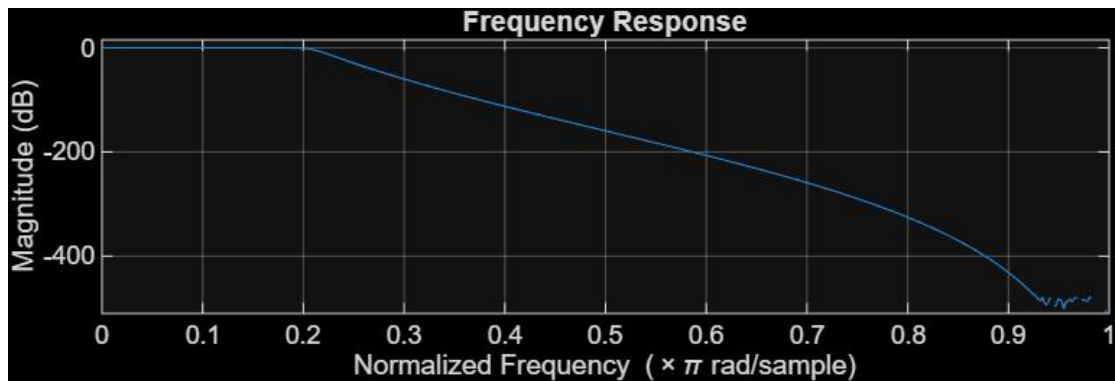
```

Command Window

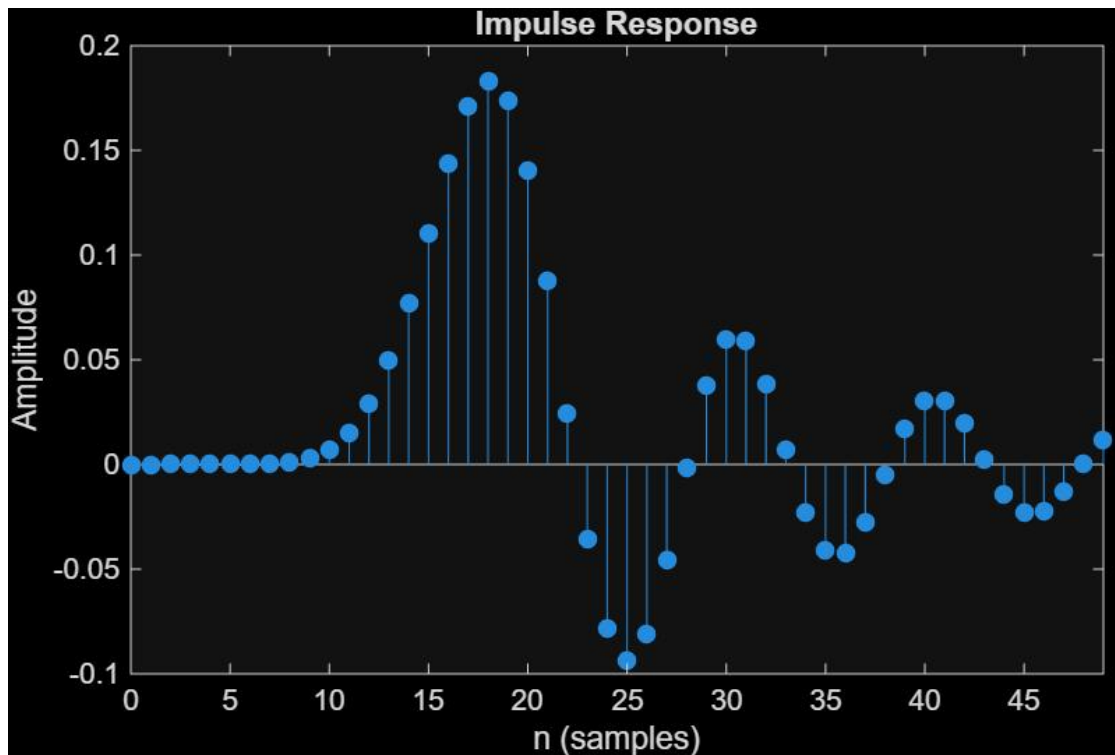
Minimum Filter Order (N) = 17
Cutoff Frequency (Wn) = 0.2083
>> |

```

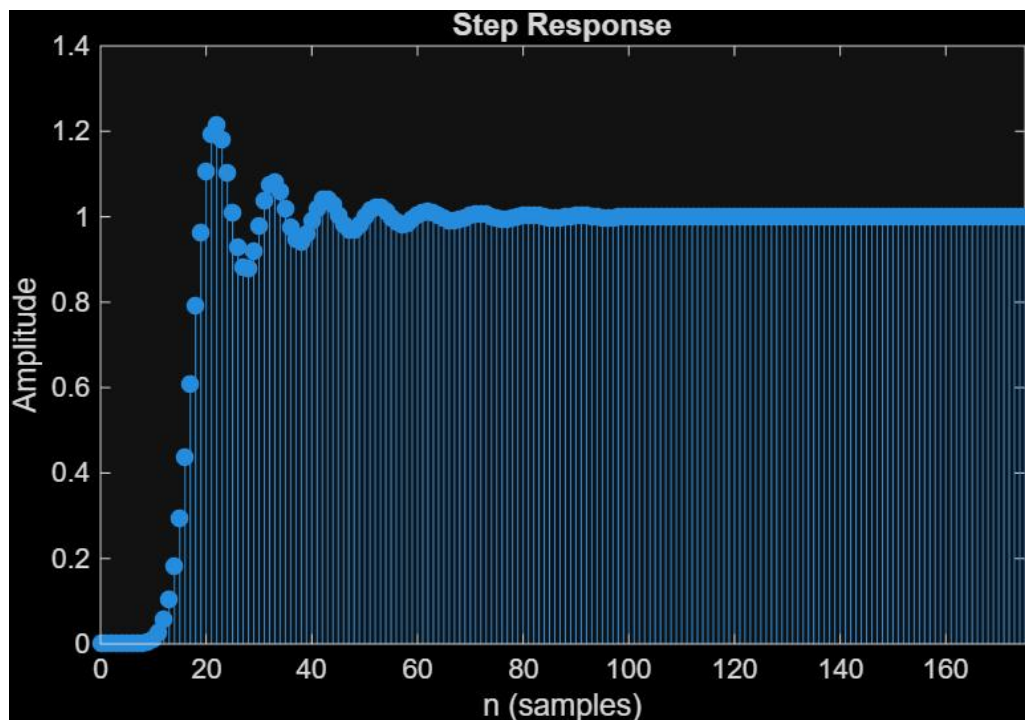
Objective 2:



Objective 3:



Objective 4:



Objective 5:

